Research Article

The Cost of Self-Service Technology Convenience: Are We Designing Human Connection Out of Our Experiences?

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HF-720 Internationalization & World Ready Product Design | Professor David | October 11, 2026



In design school, we are taught over and over again that the best experiences are seamless, intuitive, and frictionless. Remove unnecessary steps, reduce cognitive load, minimize effort, and make the experience as painless as possible for the user. After all, *isn't that the goal of good experience design?*

But what happens when we start viewing the humans involved in an experience as the friction itself? When did human interaction become something we needed to design out of our everyday lives?

Perhaps this phenomenon is a result of the COVID-19 pandemic that occurred from 2019 - 2023, when interacting with other people suddenly became a potential threat to our health. Or perhaps the pandemic simply accelerated a cultural shift that had already been developing for decades. To understand how we arrived at today's increasingly contactless economy, we must first pull back the curtain and examine what our everyday experiences once involved, and how technology has been quietly changing them behind the scenes.

Long before COVID came into the picture, human interaction and connection were expected parts of many of our everyday experiences. The cashier or pharmacist at your local grocery store knowing your name and remembering what you last told them made the experience feel familiar and personal. The waitress recognizing your family at your favorite restaurant, the barista remembering your coffee order, or the mail carrier stopping to chat. These seemingly insignificant interactions helped create a sense of belonging and community. Contributing to the societal threads we once knew, we were cognizant of the humans part of our experiences because we interacted with them to complete daily tasks. It wasn't always the product or service that made an experience memorable; often, it was the people behind it. There was a social thread binding us together through these everyday interactions, one that extended beyond the transaction itself.

Today, a growing number of our everyday experiences are facilitated by self-service technologies (SSTs), which allow consumers to access products and services independently, without requiring direct assistance from another person. These technologies have become so integrated into our daily routines that we rarely stop to think about how much they've changed the way we interact with businesses and the people behind them.

Of course, not every interaction was meaningful or enjoyable, and self-service technology is hardly a new concept. ATMs, online shopping, and automated services

existed long before the pandemic. But historically, interacting with another person was often simply part of the experience, whether you wanted it to be or not. Today, that interaction is increasingly becoming optional, and in some cases, eliminated altogether.

In fact, businesses and researchers were already exploring this transition long before we ever heard the words "social distancing." Meuter and colleagues (2000) studied self-service technologies and how they were changing the relationship between customers and businesses. Their research examined more than 800 customer experiences involving self-service systems, identifying what made consumers satisfied or dissatisfied when technology replaced traditional employee-assisted services. This was more than two decades before COVID, yet businesses were already investigating how to deliver services without requiring the same level of employee involvement.

Then came the expansion of e-commerce, smartphones, digital platforms, and mainstream internet. Shopping no longer required a trip to the store, and paying for a service no longer required speaking to an employee. Technology increasingly allowed consumers to complete tasks independently, often faster than ever before. These innovations genuinely improved experiences for the consumer, making them more accessible and convenient. But they also began changing the role that human interaction played in these everyday transactions.

Shopping for clothes, household items, and even groceries from the comfort of our couches has become a normal part of modern life. We've accepted that we can now order dinner without ever having to speak to a restaurant employee, check out at a grocery store without interacting with the cashier, and in 2026, we can even take a ride in a self-driving car without a driver present to have awkward small chat with. What once required interaction and connection with another human being can now be completed with a few taps on a screen right from our couch.

As we increasingly move toward a contactless economy, has anyone stopped to ask why? Why are we designing human beings out of experiences that historically involved other people, and what might we as a greater society lose as a result?

Perhaps more importantly, who exactly is driving this shift anyway? Are consumers increasingly demanding experiences without human interaction, or are businesses and stakeholder decisions shaping our expectations by designing experiences that make human involvement unnecessary?

Research behind this very topic suggests that Americans' preferences and social behaviors have been changing for quite some time now. A study by Twenge and Spitzberg (2020), analyzing data from more than 191,000 Americans, found that time spent in non-digital social interactions declined significantly between 2003 and 2017, particularly among adolescents and young adults. These findings suggest that changes in how Americans socialize were already in motion well before the COVID-19 pandemic and its subsequent lockdowns changed our world. While this research doesn't establish that technology caused the decline, it raises questions about how our increasingly digital lifestyles may be influencing the way we interact with one another.

As designers and human factors experts, we must not only recognize these shifts in human behavior but also understand why they're happening. Without understanding the driving forces behind these changes, how can we effectively design experiences that truly meet the needs of the people we're designing for? More importantly, we must ask ourselves what role we, as designers, may be playing in shaping these behaviors in the first place.

COVID certainly accelerated our adoption of contactless experiences. Suddenly, ordering groceries online, avoiding crowded stores, and minimizing interactions with strangers weren't simply conveniences; they were precautions for our health. Services like Instacart and DoorDash became increasingly integrated into our everyday routines. Suddenly, our most mundane tasks, such as grocery shopping or picking up takeout, could be completed without the time and effort they once required. But what began as a necessity for many people continued long after restrictions were lifted. In a study of 1,419 New Jersey residents, Younes and colleagues (2022) found that online grocery shopping increased during the pandemic, with many respondents expecting to continue using these services afterward.

While COVID may have certainly accelerated our adoption of contactless experiences, the shift toward self-service technology and changing consumer behaviors was already well underway. But this raises an even bigger question: Who exactly is driving this shift, and why?

Generational differences may offer another piece of the puzzle. Younger generations have grown up in a world where digital technology is integrated into nearly every aspect of their lives, potentially shaping different expectations and mental models around communication, convenience, and social interaction. A 2025 GoDaddy survey of 1,500 American consumers found that 54% of Gen Z and 50% of Millennials preferred shopping methods that allowed them to avoid human interaction, compared with 29% of Gen X and Baby Boomers. The same survey found that younger consumers were also more likely to prefer AI chatbots over human customer service representatives, although this preference was considerably less common. These findings suggest that younger consumers may be more comfortable with, or even prefer, experiences that minimize interpersonal interaction.

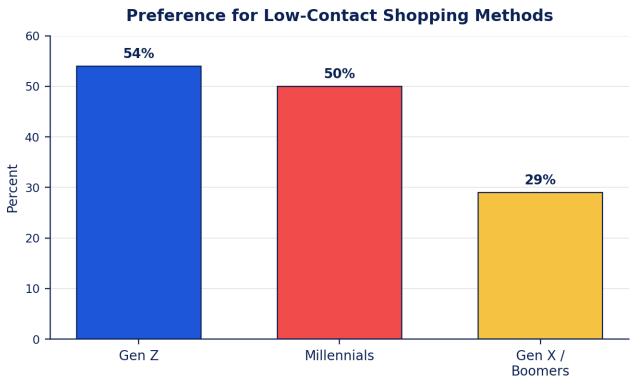


Figure 1. Preference for low-contact shopping methods by generation. Data source: GoDaddy Consumer Pulse (2025).

But again, why? Is it convenience, efficiency, independence, social anxiety, or simply a different understanding of what a good experience should look like? And are younger generations truly driving this change alone, or are they adapting to a world that has increasingly been designed around it?

Perhaps the more important question isn't whether technology has made our experiences easier. Clearly, it has. The question is whether, in our pursuit of seamless and frictionless design, we've started treating human connection as an inconvenience rather than recognizing it as something that contributes value to the experience itself.

When Did Human Interaction Become the Inconvenience?

To understand why modern consumers are increasingly gravitating toward contactless experiences, we must look at what motivates them to use these self-service technologies in the first place. Convenience is an obvious answer. Why wait in line at a grocery store when you can scan your own items and leave? Or better yet, have someone else do all the shopping for you and leave the precious cargo right at your doorstep. Why call a restaurant when you can place your order through an app and have someone else do the hard work for you? Why put up with awkward small talk with a driver when a self-driving vehicle can take you where you need to go, and even play the music that you like?

The appeal of self-service technology is understandable. These technologies give consumers greater independence and control over their experiences. But the convenience they provide isn't necessarily limited to saving time. For some people, avoiding the interaction itself is part of the appeal. Which brings us to the elephant in the room: At what point did interacting with other people become something we increasingly seek to avoid rather than embrace? Aren't we, as humans, social beings after all?

The 2025 GoDaddy survey demonstrates that this preference exists among a substantial portion of younger American consumers. However, a preference for avoiding interaction doesn't necessarily indicate alone that someone dislikes people or struggles with socialization. There may be countless reasons why someone chooses an automated experience, including convenience, privacy, social anxiety, disabilities, or simply wanting to complete a task without interruption.

The more interesting question posed here is whether these preferences are something consumers have always had, or something we're learning through repeated exposure to technology, reinforced by American cultural values of independence and autonomy (Markus & Kitayama, 1991).

Consider how quickly our expectations adapt when a better or more convenient experience becomes available. A service that once felt revolutionary eventually becomes ordinary. Two-day shipping was once an exceptional benefit, but today many consumers expect their purchases to arrive within days without an extra cost. AlixPartners' (2024) U.S. home-delivery research found that consumers' expectations of an acceptable delivery window shortened from approximately 5.5 days in 2012 to 3.5 days in 2024.

This doesn't necessarily mean Americans have become less patient as people. But it does demonstrate how modern

consumer expectations can change as service experiences evolve to disintegrate friction.

Could the same thing be happening with human interaction? As businesses increasingly offer experiences where we don't need to interact with another person, are we becoming more accustomed to, and perhaps more expectant of, those contactless experiences?

Research by Giebelhausen and colleagues (2014) offers an interesting perspective. Their study examined how technology influences interactions between customers and service employees. They found that when employees established a positive connection with customers, introducing technology could interrupt that interaction and negatively affect the customer's evaluation of the experience. However, when an employee had not established rapport, technology could improve the experience by allowing the consumer to avoid a potentially unpleasant interaction.

In other words, technology wasn't necessarily what made the experience better or worse. The human interaction it replaced mattered.

This finding challenges the assumption that removing an employee from a service automatically improves the customer experience. Sometimes the employee is the obstacle, but sometimes the employee is the very reason the experience is positive.

So why do we increasingly design experiences today as though the interaction itself is something to eliminate?

Are Consumers Driving the Change, or Are Businesses?

Perhaps we've been asking the wrong question here all along. Rather than focusing entirely on why consumers prefer contactless experiences, we should also consider why businesses are so eager to provide them.

Self-service technologies can offer businesses several advantages. Automated systems can increase transaction capacity, standardize processes, reduce certain operating costs, and allow employees to focus on other responsibilities. From a business perspective, these are attractive benefits to consider.

Research by Bitner and colleagues (2002) was already examining how businesses could successfully introduce self-service technologies and encourage consumers to adopt them. Their work emphasized that technology adoption requires more than simply making a system available; consumers must perceive a benefit and have an experience that encourages continued use.

This introduces an interesting relationship between what businesses offer and what consumers come to expect.

If a business introduces self-checkout because it improves operational efficiency, and consumers eventually become accustomed to using it, was that change originally driven by consumer demand? Or did the business introduce a new way of completing the experience that consumers eventually accepted as the improved new “normal”?

It's likely a combination of both. Businesses respond to consumer preferences, but the experiences they create can also influence those very preferences. And when those technologies become standard, they become the new expectation. Consumers may then have fewer opportunities to choose traditional human-assisted services, even when they prefer them.

Consider a grocery store that reduces its cashier-assisted lanes in favor of self-checkout. A shopper who prefers interacting with a cashier may technically still have that option, but if the remaining cashier-assisted line has a longer wait, the choice becomes less straightforward. Is the consumer choosing self-checkout because they prefer it, or because the experience has been designed to make it the most convenient option?

This is where experience design becomes particularly influential to consumer behavior and their ensuing choices. We aren't simply creating interfaces, services, or experiences that respond to existing human behaviors. We may also be creating the environments that shape those behaviors.

And if businesses increasingly optimize experiences around efficiency, automation, and reducing employee involvement, we need to consider whether we're measuring success only from the perspective of the individual transaction rather than the broader societal experience.

What Are We Losing When We Remove Human Interaction?

This brings us back to the familiar cashier, waitress, barista, and mail carrier. Those interactions might seem

small or insignificant, but research suggests they can contribute to our sense of belonging and well-being.

Sandstrom and Dunn (2014) conducted a study examining whether brief interactions with strangers could positively affect people's emotional experiences. In one experiment, customers who engaged in a genuine social interaction with a barista, such as smiling, making eye contact, and having a brief conversation, reported more positive feelings than those who completed their purchases as efficiently as possible.

Think about that for a moment. The very interaction that might now be deemed unnecessary from an efficiency standpoint actually contributed something positive to the experience.

The researchers also investigated the importance of what psychologists call "weak ties," or social relationships with acquaintances and people outside our closest social circles. In a separate study, Sandstrom and Dunn (2014) found that interactions with these weaker social connections were associated with greater feelings of happiness and belonging.

These are the people we recognize but don't necessarily know well. The cashier we see every Saturday, the neighbor we occasionally speak to, or the barista who remembers our usual order. We might not consider them important relationships, but these interactions are what help connect us to the local communities around us.

Now imagine removing more and more of those opportunities and human interactions from everyday life.

Instead of visiting the grocery store, we order delivery. Instead of speaking with a barista, we place an order through an app and pick it up without conversation. Instead of interacting with a driver, we take a self-driving vehicle. Though each individual experience may become more convenient, collectively, we're also reducing opportunities for the small interactions that once took place naturally throughout our day.

This doesn't mean everyone who uses these services will become lonely or socially disconnected. The research we have today doesn't establish that conclusion. But it instead raises another important question about whether the cumulative removal of these everyday small interactions could have larger consequences that aren't immediately visible or felt.

After all, we don't necessarily recognize the value of these small encounters until we begin considering what life might look like without them. And perhaps that's where the true cost of convenience becomes more difficult to measure.

But Isn't Avoiding Human Interaction Sometimes a Good Thing?

Before we start blaming technology for the decline of society, we should acknowledge that contactless experiences provide genuine benefits, and for some people, those benefits are significant.

Someone with a disability may find grocery delivery far more accessible than navigating a crowded store. A person experiencing social anxiety may appreciate the independence of ordering food without speaking to an employee. Someone working multiple jobs may rely on delivery services to save time, while an individual living in a rural area may benefit greatly from fast-delivery access to products that aren't available locally.

In these situations, eliminating certain interactions isn't inherently removing something valuable. It may simply be removing a barrier that previously made the experience difficult or inaccessible. Research suggests that some consumers don't necessarily want human interaction removed from every experience. Rather, for some, human interaction may be the part of the experience left on the chopping block when other needs, such as accessibility and independence, take priority.

Benoit and colleagues (2024) examined consumer preferences for autonomous retail stores, such as Amazon Go, where customers can shop and complete purchases without interacting with any employees. Their research found that consumers generally preferred having staff available, even when certain automated features were attractive. In particular, access to assistance and the ability to verify purchases influenced how consumers evaluated autonomous shopping experiences.

More recently, Gartner (2026) reported that 87% of surveyed customers considered access to a human representative essential when companies use generative AI in customer service. These findings suggest that consumers may appreciate the convenience of automation while still wanting access to human assistance when needed.

Perhaps the problem isn't that we're giving consumers the ability to avoid interaction. Perhaps the real problem begins when we design experiences that remove the opportunity for interaction altogether.

There's an important difference between offering self-checkout as an alternative and replacing every human cashier with a machine counterpart. One creates flexibility and choice, while the other makes a decision on behalf of every consumer, regardless of their needs or preferences.

If we're truly designing experiences around people, shouldn't we be considering both?

The Hidden Costs of an Increasingly Contactless Society

The potential consequences of contactless experiences also extend beyond our social interactions. As consumers become more accustomed to automated services, businesses may become increasingly dependent on digital systems, customer data, and technologies that replace or restructure human labor.

Consider what happens when a traditional in-person customer interaction becomes a digital transaction. Instead of simply purchasing an item from an employee, a consumer may use a mobile application that records their location, purchase history, preferences, and other behavioral information. The convenience of the experience may come with a less visible exchange of personal information for the consumer.

Pew Research Center (2023) found that 81% of Americans were concerned about how companies use their personal data, while 73% felt they had little or no control over the information companies collected about them. This suggests that consumers aren't necessarily comfortable with extensive data collection, even as they continue participating in digital experiences that depend on it.

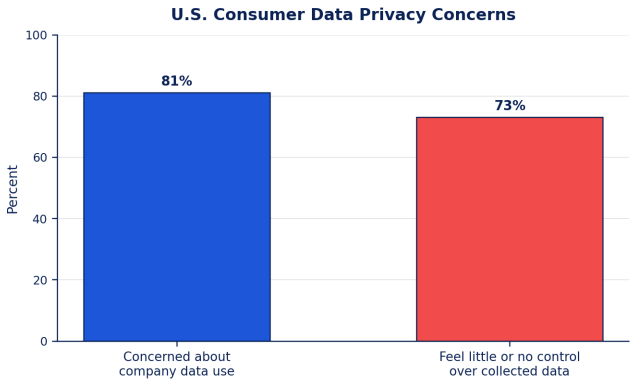


Figure 2. U.S. consumer data privacy concerns. Data source: Pew Research Center (2023).

From an experience design perspective, this raises another concern. Are consumers truly making informed decisions about the technologies they use, or have data collection and digital tracking become so deeply integrated into everyday services that avoiding them is increasingly difficult?

The economic consequences attributing here deserve consideration as well. When businesses automate transactions, the work previously performed by employees doesn't always disappear. Sometimes it shifts to customers, who now scan their own groceries, enter their own orders, troubleshoot automated systems, or navigate digital customer support.

Other forms of automation can change staffing requirements and the responsibilities of workers who remain. While these changes may improve business efficiency, they also raise questions about who benefits most from the redesign of an experience and who bears its costs.

We often discuss convenience from the consumer's perspective, but the experience doesn't exist in isolation. There's a much larger ecosystem of people, services, and interactions to consider. Employees, communities, and even people who never directly use the technology can be affected by these design decisions.

Perhaps this is one of the biggest limitations of evaluating experiences primarily through usability and efficiency. We can measure how quickly someone completes a transaction, how many steps we've removed, and whether users report satisfaction. But those measurements don't necessarily tell us what the broader consequences of the experience might be.

An experience can be successful for the person using it while still creating problems elsewhere in the system.

What Does This Mean for the Future of Experience Design?

As experience designers, we have an opportunity to question whether the principles we've been taught should always be applied in the same way. As consumer behaviors and expectations continue to evolve, we may find ourselves revisiting these same questions time and time again. And that's not a bad thing; it's something we should embrace as part of our responsibility as designers.

Reducing cognitive load, improving accessibility, and eliminating unnecessary steps are valuable design practices. The intention isn't to make experiences more difficult or require consumers to participate in social interactions they don't want. But perhaps we need to become more intentional about distinguishing between friction that prevents people from accomplishing their goals, and interactions that contribute meaningfully to the experience.

Not all friction is necessarily bad, and not everything that takes additional time is inherently inefficient.

A conversation with a cashier might make a transaction take an additional minute, but if that conversation contributes to someone's sense of community belonging, is it really a negative part of the experience? If a customer values being recognized by an employee, should that relationship be considered something to optimize away?

And if some consumers prefer automated experiences while others value human interaction, shouldn't good experience design account for both?

The solution may not be to abandon automation, but to reconsider how we implement it.

Instead of designing experiences that completely eliminate human involvement, businesses could offer meaningful choices between automated and employee-assisted services. Technology could handle repetitive tasks while employees remain available for moments that require empathy, reassurance, problem-solving, or personal connection.

We could also reconsider how we evaluate the success of these experiences. Rather than measuring efficiency and satisfaction alone, designers could investigate whether an experience preserves consumer autonomy, supports accessibility, builds trust, and considers the social needs of the communities it serves. These questions are particularly important as artificial intelligence and increasingly autonomous technologies become integrated into more aspects of everyday human life.

If this phenomenon continues, future generations may grow up in a world where human interaction is no longer an expected part of many everyday experiences, but rather an option that technology has made redundant. Whether that fundamentally changes how people socialize, form relationships, or participate in their communities remains an open research question.

We shouldn't wait until those consequences become more apparent before asking whether the direction we're heading is the one we actually want.

As designers, we have considerable influence over how people experience the world around them. It is our responsibility to think of the impacts our design choices have on the people using them. The systems, experiences, and services we create can change more than how someone completes a task; they can influence expectations, behaviors, and opportunities for human connection.

Self-Service technology has undeniably made our lives easier in countless ways, and that progress shouldn't be dismissed. But convenience and ease alone shouldn't be the measure of a successful experience. With that, I leave you with one final thought and question to consider:

We've spent decades learning how to design friction out of everyday life. Maybe it's time we started asking whether, somewhere along the way, we've been designing out parts of experiences that was never friction in the first place.

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